

FlowPOS Demand Forecasting: Model Selection & Benchmarking

Report A

March 2026

1. Introduction

This report presents the results of a large-scale model exploration for demand forecasting in FlowPOS, a Danish restaurant and food-service POS platform. The objective was to identify the best forecasting model by evaluating multiple model families against a business-impact cost metric.

A companion report (Report B) builds on these findings with investigations into data recency, seasonality, and per-store modeling strategies. Where relevant, findings from Report B are referenced.

2. Dataset Overview

2.1 Source Tables

The dataset consists of real POS transaction data from Danish restaurants, pubs, kiosks, and food shops.

Table	Rows	Description
fct_orders	294,569	Individual orders with timestamp, store, payment, type
fct_order_items	481,624	Line items: item, quantity, price, cost
dim_items	88,921	Menu items with titles, prices, categories
dim_places	2,091	Store metadata and configuration
dim_campaigns	642	Promotional campaigns
dim_users	22,956	Customer records

2.2 Key Characteristics

The dataset spans **59 days** (December 18, 2023 to February 16, 2024) across **308 active stores** with **15,266 unique items**. Total revenue is 39.1M DKK with an average order value of 133 DKK. The business is overwhelmingly takeaway (96%) with card payments (93%) placed via app.

Top-selling items are classic Danish kiosk/pub fare: Sodavand (soda), various beers (Øl, Alm Øl), food boxes (Lille/Mellem/Stor box), and Ristet Hotdog.

2.3 The Sparsity Challenge

The single most important characteristic for model selection is the extreme sparsity:

Metric	Value
Store-item pairs	15,266
Days in dataset	61
Possible (store, item, day) entries	931,226
Entries with non-zero demand	144,761
Sparsity	84.5%

On any given day, 84.5% of store-item combinations have zero sales. This zero-inflation is the dominant modeling challenge and fundamentally determines which model families succeed or fail.

2.4 Demand Patterns

Weekly seasonality is strong: Friday is the peak day at 1.35x average, Sunday is the trough at 0.74x. The dataset also contains a massive demand trend — orders grew roughly 10x from early December (634/day in week 51) to early February (7,504/day in week 5). This trend issue is explored in depth in Report B.

3. Evaluation Methodology

3.1 Train/Test Split

A temporal split was used: all data up to February 2 for training (323,564 rows after removing early lag-incomplete rows), and the final 14 days (February 3–16) as the test set (98,476 rows). This avoids future data leakage.

3.2 Business Cost Metric

Standard forecasting metrics (MAE, RMSE, WMAPE) were computed but the primary ranking metric is **Total Business Cost in DKK**:

$$\text{Total Cost} = \text{Waste Cost} + 1.5 \times \text{Stockout Cost}$$

Where:

- **Waste Cost** = overstock units $\times$ item price $\times$ 0.30 (30% of ingredient cost lost when food is wasted)
- **Stockout Cost** = understock units $\times$ item price (full lost revenue)

- The 1.5x multiplier on stockouts reflects that missed sales damage revenue and customer loyalty more than food waste.

This metric directly measures the business impact of forecasting errors.

3.3 Feature Engineering

38 features were engineered across six categories:

- **Time** (15 features): day of week, month, quarter, cyclical sin/cos encodings, weekend/Friday/Monday flags, season
- **Lag** (6 features): 1/7/14/28-day demand lags, same-weekday last week, 4-week same-weekday average
- **Rolling** (7 features): 7/14/30-day rolling means, 7/14-day rolling standard deviations, expanding mean
- **External** (7 features): weather stubs, holiday flags, promotion flags, store open flag
- **Categorical** (2 features): label-encoded place_id and item_id
- **Interaction** (1 feature): store-weekday interaction term

3.4 Parallel Team Execution

To maximize coverage, **10 specialized agents** were deployed in parallel, each exploring a different model family with 3–4 hyperparameter configurations. This produced 37 distinct model variants in a single session.

4. Results

4.1 Full Leaderboard

The complete ranking of all 37 model variants by Total Business Cost:

Rank	Model	MAE	WMAPE	Total Cost (DKK)	Train Time
1	RF Default	1.781	84.9%	6,035,987	18s
2	ET Default	1.901	90.7%	6,210,050	13s
3	RF Tuned	1.823	86.9%	6,212,328	29s
4	ET Tuned	1.796	85.7%	6,298,901	50s
5	RF Large	1.729	82.4%	6,387,003	154s
6	CatBoost Shallow	1.773	84.6%	6,421,793	176s
7	ET Large	1.850	88.2%	6,423,460	115s
8	CatBoost Deep	1.748	83.3%	6,453,977	328s
9	CatBoost Default	1.795	85.6%	6,454,638	120s

Rank	Model	MAE	WMAPE	Total Cost (DKK)	Train Time
10	Stacking XGB	1.688	80.5%	6,594,097	794s
11	Meta Weighted ETS	1.632	77.8%	6,654,421	0s
12	Stacking Ridge	1.896	90.4%	6,786,156	684s
13	Meta SES (alpha=0.3)	1.677	80.0%	6,831,516	0s
14	Damped Trend ETS	1.684	80.3%	6,856,446	0s
15	LightGBM Default	1.439	68.6%	6,873,856	38s
16	MA7 baseline	1.716	81.8%	6,967,146	0s
17	LightGBM Sparse	1.413	67.4%	6,985,702	41s
18	LightGBM Tuned	1.457	69.5%	7,033,513	89s
19	Croston Hybrid	1.929	92.0%	7,060,510	0s
20	NaiveLastWeek	1.797	85.7%	7,308,750	0s
21	SGD Huber	1.504	71.7%	7,489,994	1s
22	Adaptive ETS	1.863	88.8%	7,528,529	1s
23	ElasticNet	2.182	104.1%	8,186,728	91s
24	SGD Eps-Insensitive	1.583	75.5%	8,412,233	2s
25	LinearSVR Regularized	1.689	80.5%	9,367,123	197s
26	LinearSVR Default	1.692	80.7%	9,388,702	725s
27-31	Croston variants	2.2-2.3	106-109%	11.2-11.5M	0s
32-36	Linear models	3.0-3.7	143-178%	11.5-12.4M	1-637s
37	Zero baseline	2.097	100.0%	13,664,654	0s

4.2 Model Family Summary

Family	Best Variant	Cost (DKK)	vs MA7	Verdict
Random Forest	RF Default	6,036K	-13.4%	Winner
Extra Trees	ET Default	6,210K	-10.9%	Strong runner-up
CatBoost	Shallow	6,422K	-7.8%	Solid
Exponential Smoothing	Weighted ETS	6,654K	-4.5%	Best zero-training option
Stacking	XGB Meta	6,594K	-5.4%	Not worth the complexity
LightGBM	Default	6,874K	-1.3%	Best raw accuracy
Croston's SVR	Hybrid	7,061K	+1.3%	Disappointing
	SGD Huber	7,490K	+7.5%	Not competitive
ElasticNet	Default	8,187K	+17.5%	Broken on sparse data
Bayesian Ridge	ARD	12,354K	+77.3%	Completely broken

5. Analysis: Why Accuracy Does Not Equal Business Value

The most important finding from this exploration is that **forecast accuracy and business value are inversely correlated** on this dataset.

Model	WMAPE	Waste (DKK)	Stockout (DKK)	Over-predict %	Total Cost
LightGBM	67.4%	695K	4,194K	48.7%	6,986K
Sparse RF	84.9%	1,616K	2,947K	54.3%	6,036K
Default					

LightGBM is nearly twice as accurate by WMAPE yet costs 15% more. The mechanism:

Random Forest's bagging creates a natural safety buffer. Each of 100 trees is trained on a bootstrap sample. With 84.5% zeros, different trees see different zero/non-zero ratios and produce varied predictions. Their average sits slightly above the true mean — a built-in tendency to over-predict.

Gradient boosting is too precise. LightGBM and XGBoost iteratively minimize prediction error, aggressively learning that 84.5% of predictions should be zero. They achieve excellent accuracy but produce expensive stockouts when demand does materialize.

Since the business metric penalizes stockouts at 1.5x, RF's slight over-prediction bias saves 1.2M DKK in stockouts while only adding 920K DKK in waste — a net gain of

nearly 1M DKK.

6. What Didn't Work and Why

Approach	Root Cause of Failure
Linear models (Ridge, Lasso, ElasticNet, Bayesian Ridge)	Cannot learn the zero-inflation pattern. They predict moderate positive values everywhere instead of mostly zeros with occasional spikes. All scored 0% forecast accuracy and 12M+ DKK cost.
Croston's method	Designed for intermittent demand, but the sparsity here is structural (most items never sell at a given store) not stochastic (demand arriving at random intervals). In tabular form with pre-computed features, Croston loses its per-series advantage.
Stacking ensemble	The meta-learner averages base predictions, diluting RF's beneficial conservative bias. Costs 794s to train for worse results than a 18s RF.
SVR / LinearSVR	Linear kernel cannot capture the zero-inflation. RBF kernel doesn't scale to 323K rows. SGD Huber was the best SVR variant but still 7.5% worse than MA7.

7. Conclusion

Random Forest with default hyperparameters (100 trees, unlimited depth) is the clear winner for this demand forecasting problem, achieving a 13.4% reduction in business cost versus the MA7 baseline.

The model's advantage stems not from sophistication but from a happy alignment between RF's bagging mechanism and the business objective: on highly sparse data, bagging naturally produces slightly conservative (over-predicting) forecasts, which is exactly what a stockout-penalizing cost function rewards.

Report B extends these findings with data recency strategies and a global/per-store blending approach that pushes total improvement to 19%.